

Microlocal Morse Theory and Truncated Microsupport

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December 2, 2025*

Preface

In these notes, We study microlocal Morse theory, which is developed by Guillermou, Kashiwara, and Schapira in [GKS12], but replace microsupports with truncated microsupprts introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KMS03].

1 Truncated Microsupport

Definition 1.1. (i) Let $F \in D^b(\mathbf{k}_X)$. The closed conic subset $SS_k(F)$ of T^*X is defined by: $p \notin SS_k(F)$ if and only if F satisfies the equivalent conditions in Prop.

(ii) aa

2 Results

Proposition 2.1. Let F in $D^b(\mathbf{k}_M)$, and $\psi: M \rightarrow \mathbf{R}$ be a real-valued function of class C^1 , proper on $\text{supp}(F)$. Let $a < b$ be real numbers. We assume $d\psi(x) \notin SS_k(F)$ for $a \leq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t := \psi^{-1}(]-\infty, t])$. Then, the restriction morphisms

$$H^j(M_b; F) \rightarrow H^j(M_a; F)$$

are isomorphic for each $j \leq k$.

Proof. First, we have

$$H^j(M_t; F) \cong H^j \text{R}\Gamma(]-\infty, t[; \text{R}\psi_* F)$$

for $t \in I$. Since we assume that $d\psi(x) \notin SS_k(F)$ for $a \leq \psi(x) < b$, and by the formula $SS_k(\text{R}\psi_* F) \subset \psi_\pi \psi_d^{-1} SS_k(F)$, we have

$$(H^j \text{R}\Gamma_{[a, b[} \text{R}\psi_* F)_{\psi(x)} = 0$$

*2025/03/09 first draft

for each $j \leq k$. Therefore, letting $d \in \mathbf{Z}$ be the minimum number that

$$H^d \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \neq 0,$$

by the long exact sequence

$$\begin{aligned} 0 &\rightarrow H^d \mathrm{R}\Gamma_{[a, b[}(\mathrm{R}\psi_* F) \rightarrow H^d \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F) \rightarrow H^d \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \\ &\rightarrow H^{d+1} \mathrm{R}\Gamma_{[a, b[}(\mathrm{R}\psi_* F) \rightarrow H^{d+1} \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F) \rightarrow H^{d+1} \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \\ &\rightarrow \dots \\ &\rightarrow H^k \mathrm{R}\Gamma_{[a, b[}(\mathrm{R}\psi_* F) \rightarrow H^k \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F) \rightarrow H^k \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \\ &\rightarrow H^{k+1} \mathrm{R}\Gamma_{[a, b[}(\mathrm{R}\psi_* F) \rightarrow \dots, \end{aligned}$$

we have isomorphisms

$$(H^j \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F))_{\psi(x)} \xrightarrow{\sim} (H^j \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F))_{\psi(x)}$$

for any $j \leq k$ and $a \leq \psi(x) < b$, and then,

$$H^j \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F) \xrightarrow{\sim} H^j \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \quad (2.1)$$

for all $j \leq k$. Then we can conclude that

$$H^j(M_b; F) \xrightarrow{\sim} H^j(M_a; F)$$

by applying to the equality (2.1) the functor $\Gamma(]-\infty, b[; \cdot)$ and calculating as follows:

$$\begin{aligned} \Gamma(]-\infty, b[; H^j \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F)) &\cong \Gamma_{]-\infty, b[}(\Gamma(]-\infty, b[; \mathrm{R}^j \psi_* F)) \\ &\cong \Gamma(]-\infty, b[; \mathrm{R}^j \psi_* F) \\ &\cong H^j(\psi^{-1}(]-\infty, b[); F) \\ &\cong H^j(M_b; F) \end{aligned}$$

and

$$\begin{aligned} \Gamma(]-\infty, b[; H^j \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F)) &\cong \Gamma_{]-\infty, a[}(\Gamma(]-\infty, b[; \mathrm{R}^j \psi_* F)) \\ &\cong \Gamma(]-\infty, a[; \mathrm{R}^j \psi_* F) \\ &\cong H^j(\psi^{-1}(]-\infty, a[); F) \\ &\cong H^j(M_a; F). \end{aligned}$$

□

3 Truncated Morse Inequalities

The main theorem of [ST92] is the following: we assume

$$\begin{aligned} &\psi^{-1}(]-\infty, t[) \cap \mathrm{supp}(F) \text{ is compact for any } t \in \mathbf{R}, \\ &\text{there is a finite family } \{K_j\}_{1 \leq j \leq N} \text{ of disjoint compact} \end{aligned} \quad (\text{H.1})$$

subsets in X and real numbers $\{s_j\}_{1 \leq j \leq N}$ such that

$$\pi(\Lambda_\psi \cap \Lambda) = \bigcup_j K_j, \quad \psi|_{K_j} = s_j, \quad (\text{H.2})$$

$$V_j := \mathrm{R}\Gamma(K_j; \mathrm{R}\Gamma_{\{\psi \geq s_j\}}(F)) \text{ belongs to } \mathrm{D}^b(\mathrm{Mod}^f(\mathbf{k})). \quad (\text{H.3})$$

Set

$$n_l := \sum_j \dim H^l(V_j). \quad (3.1)$$

Then

Theorem 3.1 (Generalized Morse Inequalities [ST92, Theorem 1.1]). Assume (H.1), (H.2), and (H.3). Then

- (i) $R\Gamma(X; F) \in D^b(\text{Mod}^f(\mathbf{k}))$,
- (ii) setting $b_l(X, F) = b_l(R\Gamma(X; F))$, we have the inequalities

$$b_l^*(X, F) \leq n_l^*$$

where n_l is given by (3.1).

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